

Experiment analysis step 7: randomization inference

Design-based check for the primary crossover effect

W241 Plank Study

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0.1 Purpose

This document provides a design-based randomization inference check for the main crossover effect. Under the sharp null of no treatment effect, each participant's observed paired difference could have appeared with either sign, so the randomization distribution can be approximated by random sign flips.

0.2 Load primary sample

```
paired <- sanitize_dt_ascii(fread(derived_path("paired_analysis.csv")))  
  
stopifnot(nrow(paired) == 38L)  
stopifnot(uniqueN(paired$participant_id) == nrow(paired))
```

0.3 Observed statistic

```
observed_diff <- mean(paired$diff_high_minus_low)  
  
obs_tab <- data.table(  
  quantity = c(  
    "Participants",  
    "Observed paired mean difference (high - low)"  
  ),  
  value = c(  
    nrow(paired),  
    observed_diff  
  )  
)  
obs_tab[, value := round(value, 4)]  
  
kable(obs_tab, caption = "Observed statistic for randomization inference")
```

Table 1: Observed statistic for randomization inference

quantity	value
Participants	38.0000
Observed paired mean difference (high - low)	14.3694

0.4 Bridge linear model

The randomization-inference statistic is the same paired mean difference reported in the primary analysis. To make that link explicit, we first write the observed effect as the intercept from a one-sample linear model on the participant-level differences. This does not change the estimand; it only provides the conventional regression-style coefficient, standard error, and p-value before the design-based RI calculation.

```
bridge_fit <- lm(diff_high_minus_low ~ 1, data = paired)
bridge_coef <- summary(bridge_fit)$coefficients["(Intercept)", ]
bridge_ci <- unname(confint(bridge_fit)[1, ])

bridge_tab <- data.table(
  quantity = c(
    "Intercept (mean paired effect)",
    "Standard error",
    "t statistic",
    "Conventional two-sided p-value",
    "95% CI lower",
    "95% CI upper"
  ),
  value = c(
    unname(bridge_coef["Estimate"]),
    unname(bridge_coef["Std. Error"]),
    unname(bridge_coef["t value"]),
    unname(bridge_coef["Pr(>|t|)"]),
    bridge_ci[1],
    bridge_ci[2]
  )
)
bridge_tab[, value := round(value, 4)]

kable(bridge_tab, caption = "Bridge linear model for the paired estimand")
```

Table 2: Bridge linear model for the paired estimand

quantity	value
Intercept (mean paired effect)	14.3694
Standard error	4.6916
t statistic	3.0628
Conventional two-sided p-value	0.0041
95% CI lower	4.8633
95% CI upper	23.8756

0.5 Randomization distribution

```
set.seed(241)
n_sims <- 10000L
diffs <- paired$diff_high_minus_low

ri_dist <- replicate(
  n_sims,
  mean(diffs * sample(c(-1, 1), length(diffs), replace = TRUE))
)

ri_two_sided_p <- mean(abs(ri_dist) >= abs(observed_diff))
ri_one_sided_p <- mean(ri_dist >= observed_diff)

ri_tab <- data.table(
  quantity = c(
    "Simulations",
    "Observed paired mean difference",
    "Two-sided RI p-value",
    "One-sided RI p-value (high > low)"
  ),
  value = c(
    n_sims,
    observed_diff,
    ri_two_sided_p,
    ri_one_sided_p
  )
)
ri_tab[, value := round(value, 4)]

kable(ri_tab, caption = "Randomization inference summary")
```

Table 3: Randomization inference summary

quantity	value
Simulations	10000.0000
Observed paired mean difference	14.3694
Two-sided RI p-value	0.0031
One-sided RI p-value (high > low)	0.0014

0.6 Figure

```
ri_plot <- data.table(statistic = ri_dist)

ggplot(ri_plot, aes(x = statistic)) +
  geom_histogram(bins = 40, fill = "#9ecae1", color = "white", linewidth = 0.2) +
  geom_vline(xintercept = observed_diff, color = "#d94801", linewidth = 0.9) +
  geom_vline(xintercept = -observed_diff, color = "#d94801", linewidth = 0.9, linetype = "dashed") +
  labs(
    title = "Randomization distribution under the sharp null",
    subtitle = "Solid orange = observed statistic; dashed orange = symmetric reference",
    x = "Paired mean difference (high - low)",
```

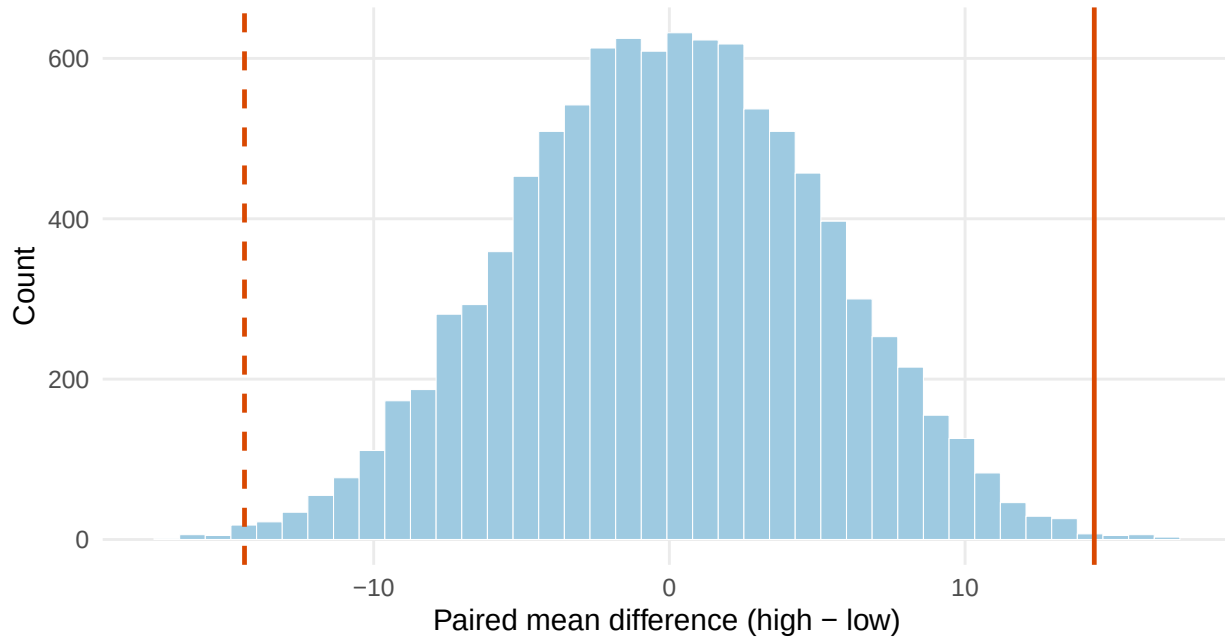
```

y = "Count"
) +
theme_minimal(base_size = 12) +
theme(panel.grid.minor = element_blank())

```

Randomization distribution under the sharp null

Solid orange = observed statistic; dashed orange = symmetric reference



0.7 Verification

```

stopifnot(length(ri_dist) == n_sims)
stopifnot(ri_two_sided_p >= 0 && ri_two_sided_p <= 1)
stopifnot(ri_one_sided_p >= 0 && ri_one_sided_p <= 1)
stopifnot(all.equal(unname(coef(bridge_fit)[1]), observed_diff, tolerance = 1e-10))

message("Observed paired mean difference: ", round(observed_diff, 3))
## Observed paired mean difference: 14.369
message("Bridge linear-model p-value: ", round(unname(bridge_coef["Pr(>|t|)"]), 4))
## Bridge linear-model p-value: 0.0041
message("Two-sided RI p-value: ", round(ri_two_sided_p, 4))
## Two-sided RI p-value: 0.0031
message("One-sided RI p-value: ", round(ri_one_sided_p, 4))
## One-sided RI p-value: 0.0014

```